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Supplemental Information

Refining chimeric antigen receptors via barcoded protein domain combination pooled screening

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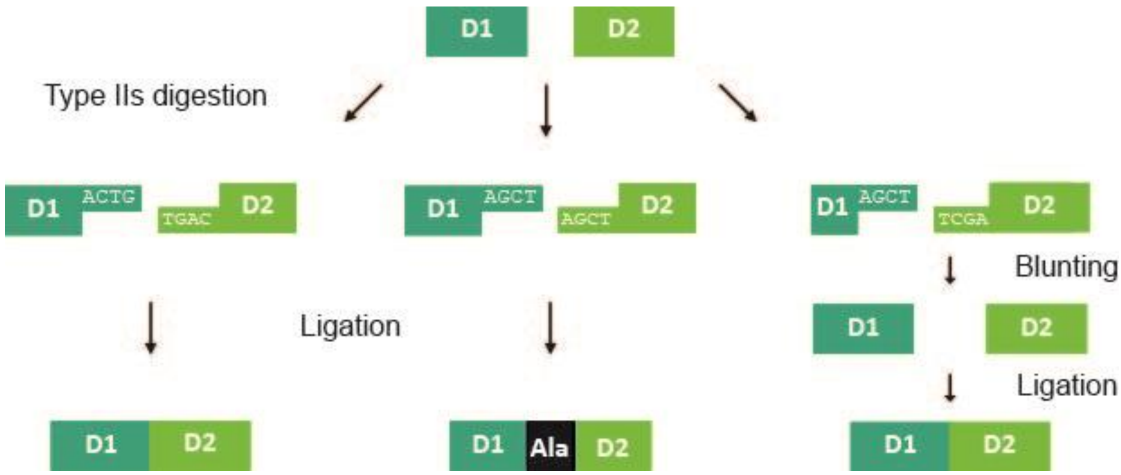
A


D1
D2
Type IIs digestion
D1 ACTG
TGAC D2
D1 AGCT
AGCT D2
D1 AGCT
TCGA D2
Blunting
D1
D2
Ligation
Ligation
D1 D2
D1 Ala D2
D1 D2

B

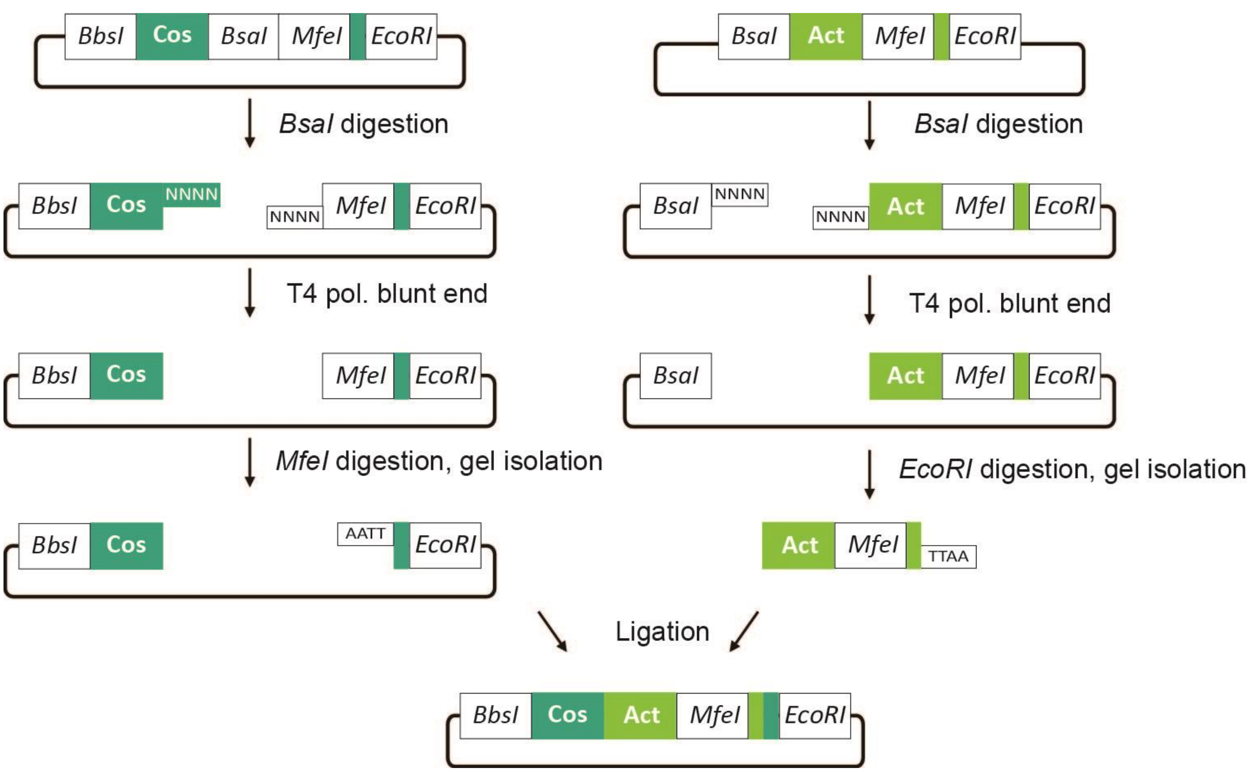

BbsI Cos BsaI MfeI EcoRI
BsaI Act MfeI EcoRI
BsaI digestion
BsaI digestion
BbsI Cos NNNN
NNNN MfeI EcoRI
BsaI NNNN
NNNN Act MfeI EcoRI
T4 pol. blunt end
T4 pol. blunt end
BbsI Cos
MfeI EcoRI
BsaI
Act MfeI EcoRI
MfeI digestion, gel isolation
EcoRI digestion, gel isolation
BbsI Cos
AATT EcoRI
Act MfeI TTAA
Ligation
BbsI Cos Act MfeI EcoRI

Figure S1. Combinatorial Serial Assembly of barcoded CAR domains.

(A) Type IIs restriction enzymes can be used to seamlessly fuse protein-coding sequences as they cut a distance away from their recognition sites. In the simplest case (left), the overhang sites can be generated at amino acid residues shared between protein coding segments, as when mutagenizing a single protein or closely related protein family. In cases where the protein sequences to be fused do not have compatible overhangs, small linkers such as 1-2 alanine residues can be added (middle). Alternatively, the overhangs can be designed so they can be blunted by T4 polymerase (right), while keeping the coding sequences unchanged. The fragments can be directionally cloned in a blunt/sticky end hybrid ligation reaction. (B) Sub-pools were generated for each CAR domain (scFv, hinge, transmembrane, costimulatory and activating domain) by cloning each into an entry vector with a barcode sequence, one barcode per individual domain. For the assembly of two subpools, for example the costimulatory and activating domains, each was digested with a type IIs restriction enzyme, blunt-end fixed, digested by either *MfeI* or *EcoRI* then ligated in a hybrid sticky-end/blunt reaction. Dual antibiotic selection improves generation of desired ligation product. This allows seamless assembly of chimeric protein domains independently of the type IIs overhangs. To make a third generation CAR library, the costimulatory domain subpool was assembled twice.

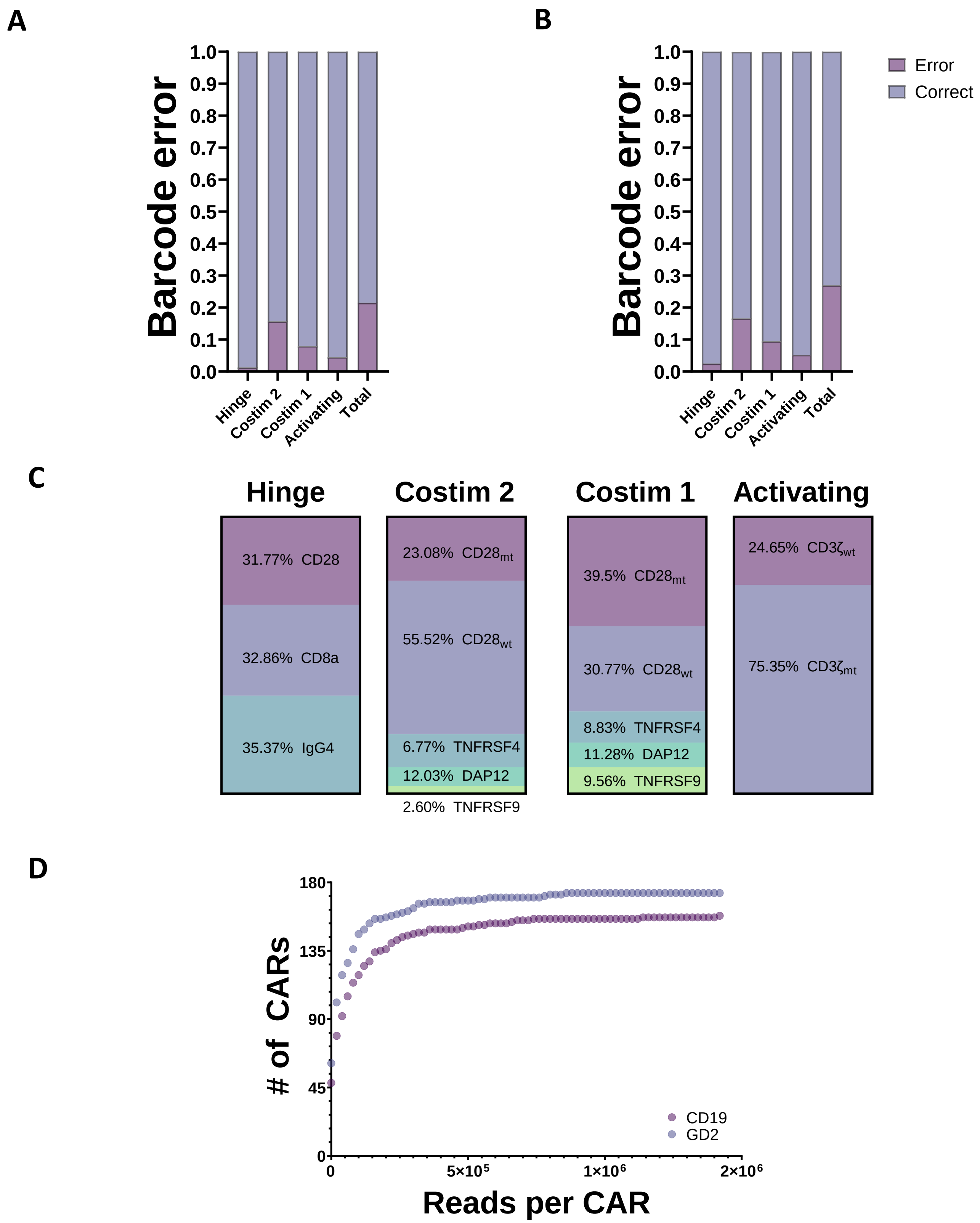


Figure S2. Validation of assembled barcoded CAR library.

(A-B), Barcoding accuracy was determined by comparing CAR coding sequence to its corresponding barcode via Sanger sequencing of CAR library from plasmid clones (A, n=79), Jurkat cell genomic DNA PCRs (B, n= 43). (C) CAR domain frequencies in the library show up to 2-10-fold differences between domains within a subpool, predominantly in the costimulatory domains. (D) Cumulative distribution of CD19 and GD2-CAR construct barcodes reads determined by NGS, ~13 million reads per library. Fully assembled CARs constitute 95-97% of all reads.

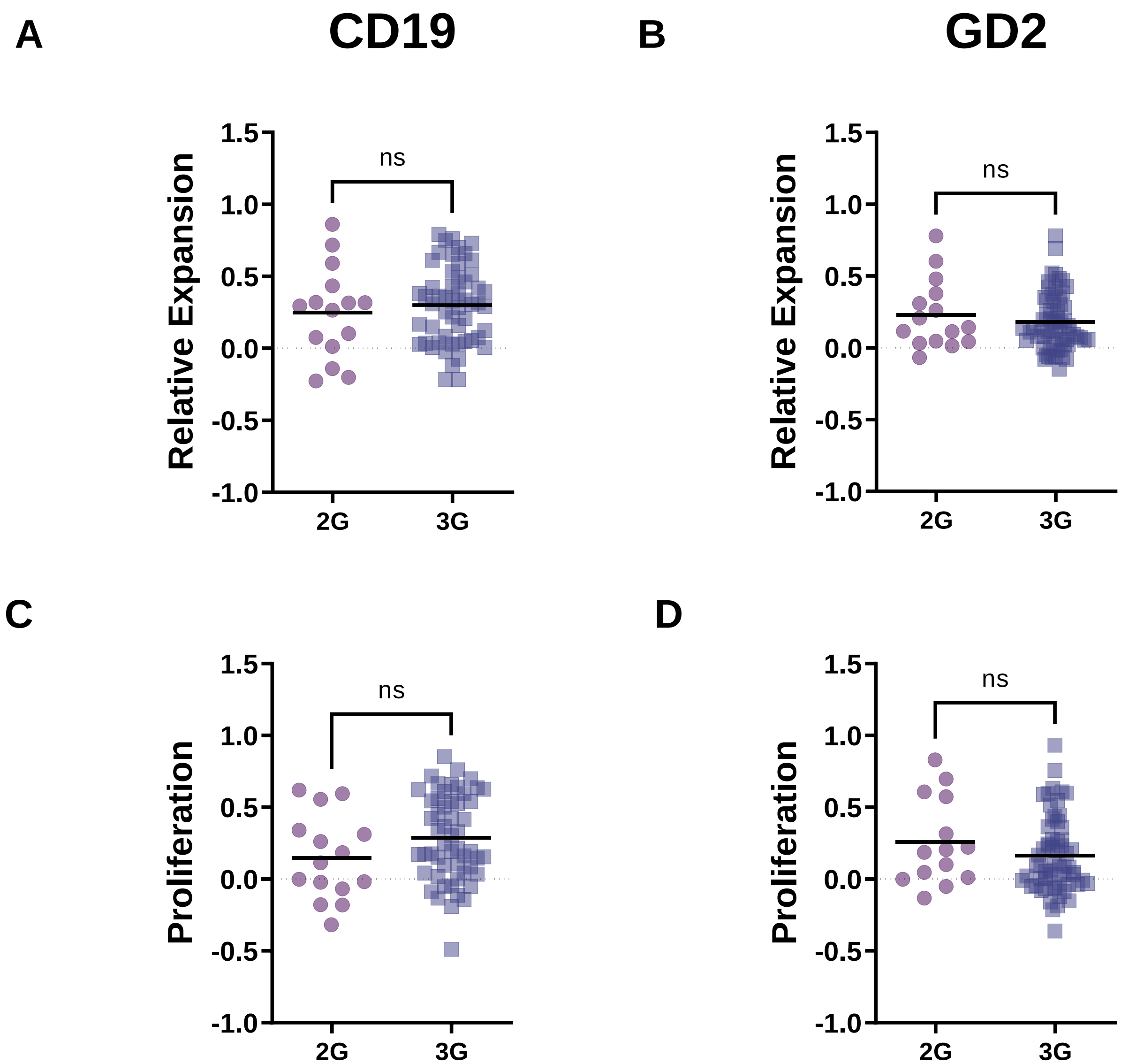


Figure S3. GD2 and CD19 expansion library screens reveal no difference between 2nd and 3rd generation CARs.
(A-D) Analysis of screened CARs with CD3ζ_{wt} grouped based on second- or third-generation costimulatory domain architecture as described in Figure 2. There is no significant difference between these CAR architectures for either CD19 or GD2-CARs (Mann-Whitney test).

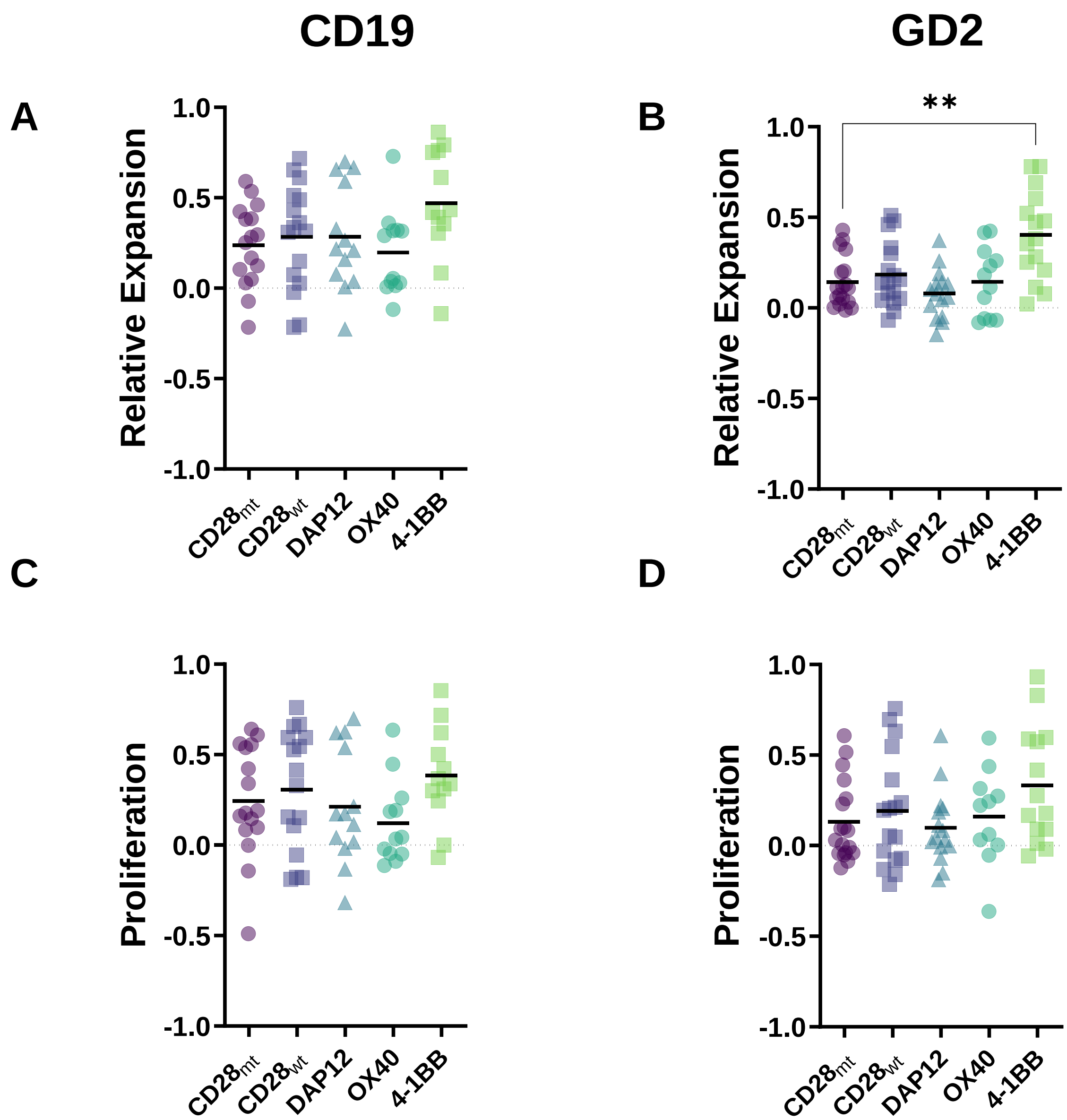


Figure S4. GD2 and CD19 library screens showed preference for 4-1BB costimulatory domain at the membrane proximal position.

(A-D) Analysis of CD3ζ_{wt} CARs costimulatory domain, relative to a $CD28_{mt}$ negative control. In the case of third–generation CARs the membrane-proximal domain was included. 4-1BB performed best, being significantly better than the $CD28_{mt}$ at mediating GD2 relative expansion (B, p= 0.0088 Kruskal-Wallis test with Dunn's multiple comparison test)

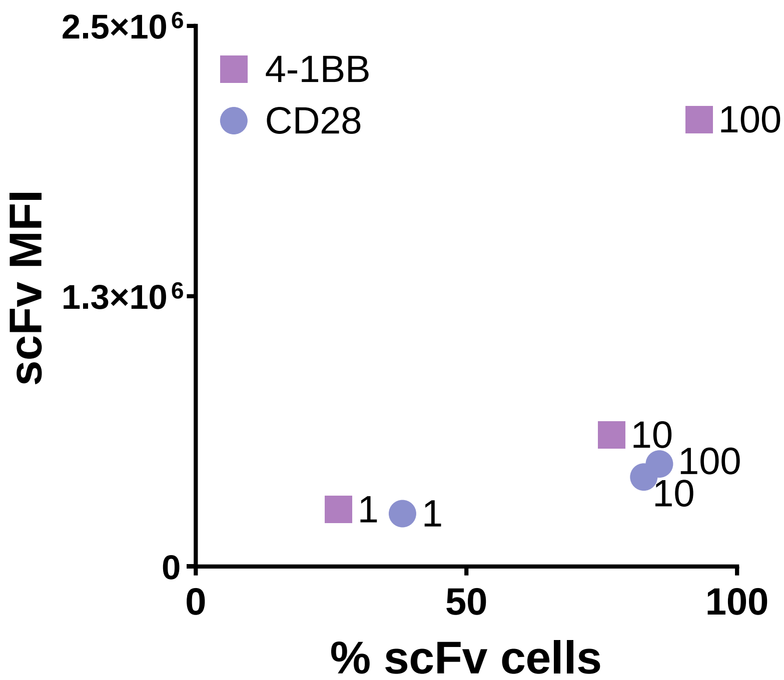


Figure S5. CD28 CAR has lower surface expression compared to 4-1BB when transduced with high virus concentrations.
GD2 4-1BB and CD28 CAR constructs were transduced with 100, 10, or 1 %v/v viral supernatant and CAR MFI/transduction rate was evaluated at 10 days post-transduction by flow cytometry. n=4 donors.

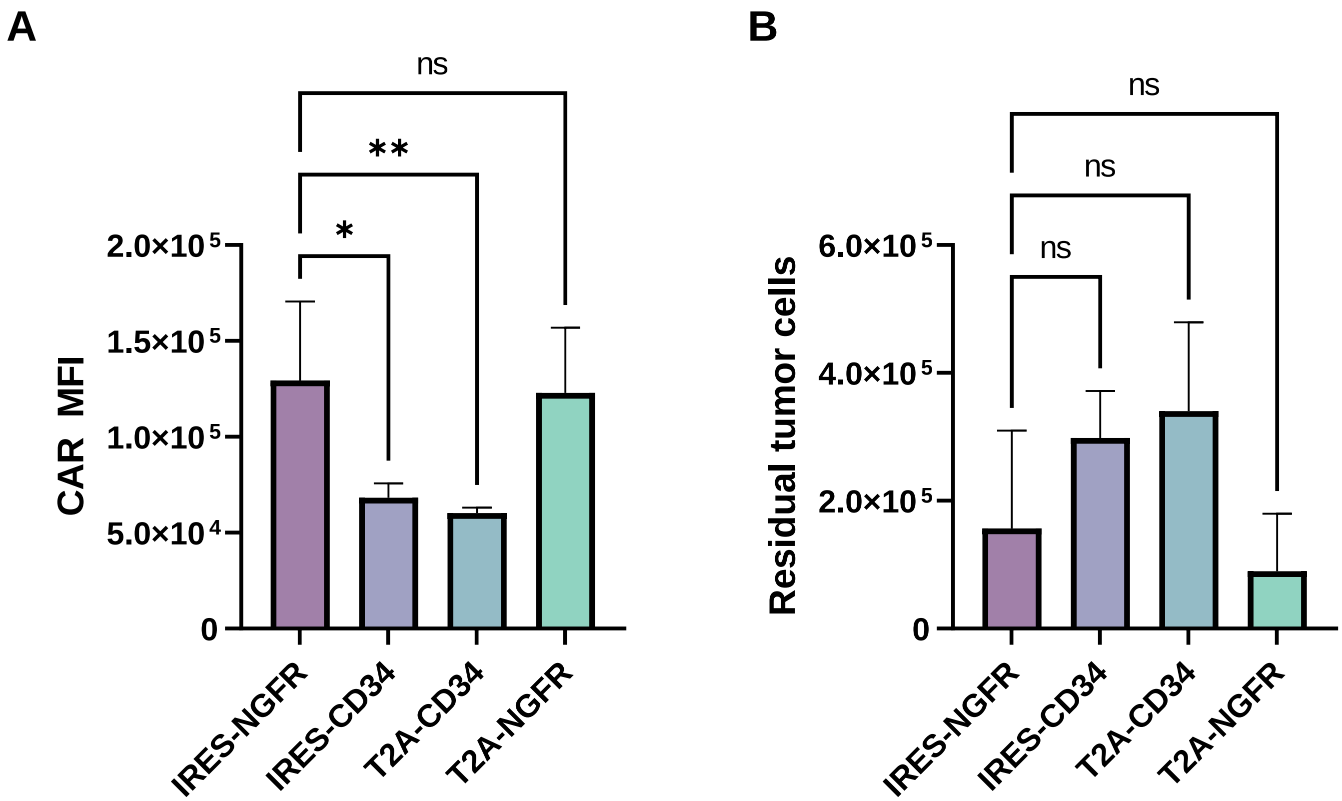


Figure S6. Role of transduction markers in GD2-CAR expression and antitumor activity. Comparison of how different transduction markers (NGFR vs CD34) and bicistronic linkers (T2A vs IRES) affect GD2-41BB CAR (A) expression, and (B) residual CHLA-255 tumor cells after four cycles of tumor antigen stimulation (relative to the IRES-NGFR used in the CAR libraries). n=4 donors, **p≤0.01; *p≤ 0.05; One-way ANOVA with Dunnett's multiple comparisons test.

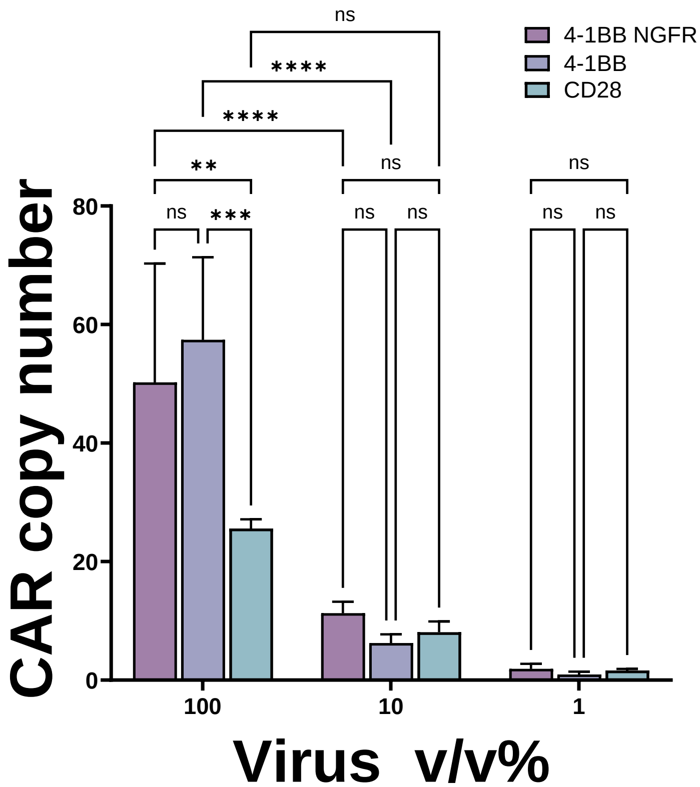
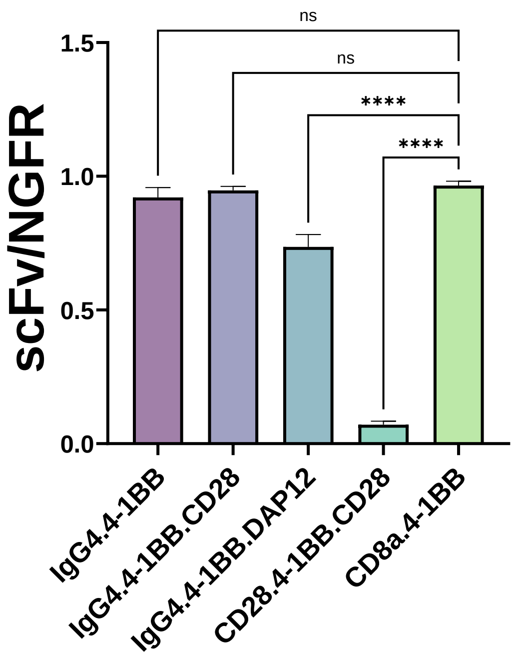


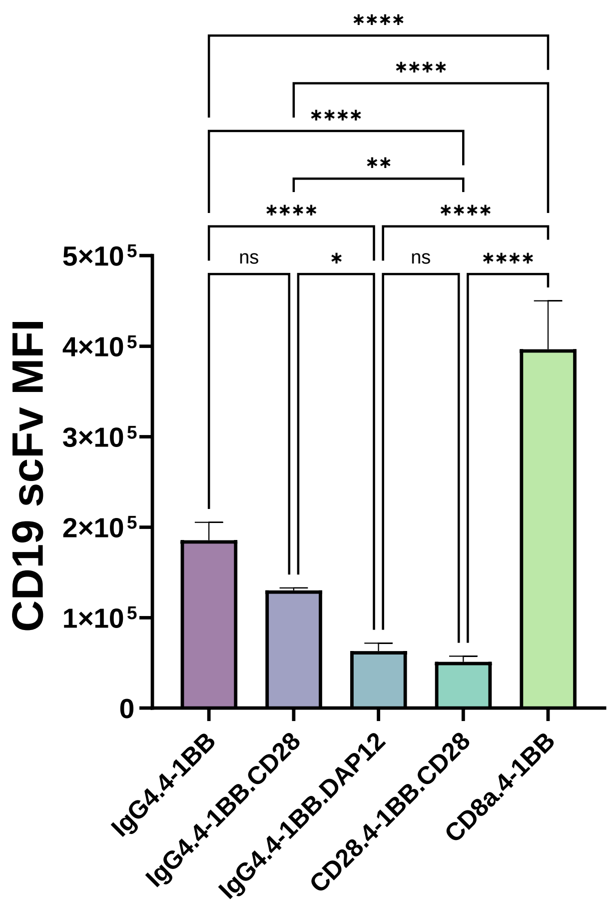
Figure S7. CAR gene copy number in transduced cells at different virus dilutions.
T cells were transduced with the different CAR constructs at varying virus v/v%, genomic DNA was extracted 10 days later, and copy number was determined by qPCR. **** p≤0.0001; *** p≤0.001; ** p≤0.01; Two-way ANOVA.

A



ns
ns
****
****
1.5
1.0
0.5
0.0
scFv/NGFR
IgG4.4-1BB
IgG4.4-1BB.CD28
IgG4.4-1BB.DAP12
CD28.4-1BB.CD28
CD8a.4-1BB

B



****
****
****
**
****
****
ns
*
ns
****
5×10^5
4×10^5
3×10^5
2×10^5
1×10^5
0
CD19 scFv MFI
IgG4.4-1BB
IgG4.4-1BB.CD28
IgG4.4-1BB.DAP12
CD28.4-1BB.CD28
CD8a.4-1BB

Figure S8. CD19-CAR constructs relative expression fraction and cell surface expression vary depending on hinge and costimulatory domains.

T cells were transduced with the different CD19-CAR constructs and 10 days later CAR and NGFR surface expression was measured via flow cytometry. (A) Fraction of transduced (NGFR+) cells expressing CARs were compared to the clinically validated CD8a.4-1BB structure; **** $p \leq 0.0001$, One-way ANOVA with Dunnett's multiple comparison. (B) CAR MFI from the CAR+ fraction was compared among CD19-CAR architectures; **** $p \leq 0.0001$, *** $p \leq 0.001$; ** $p \leq 0.01$; One-way ANOVA with Turkey's multiple comparison test.

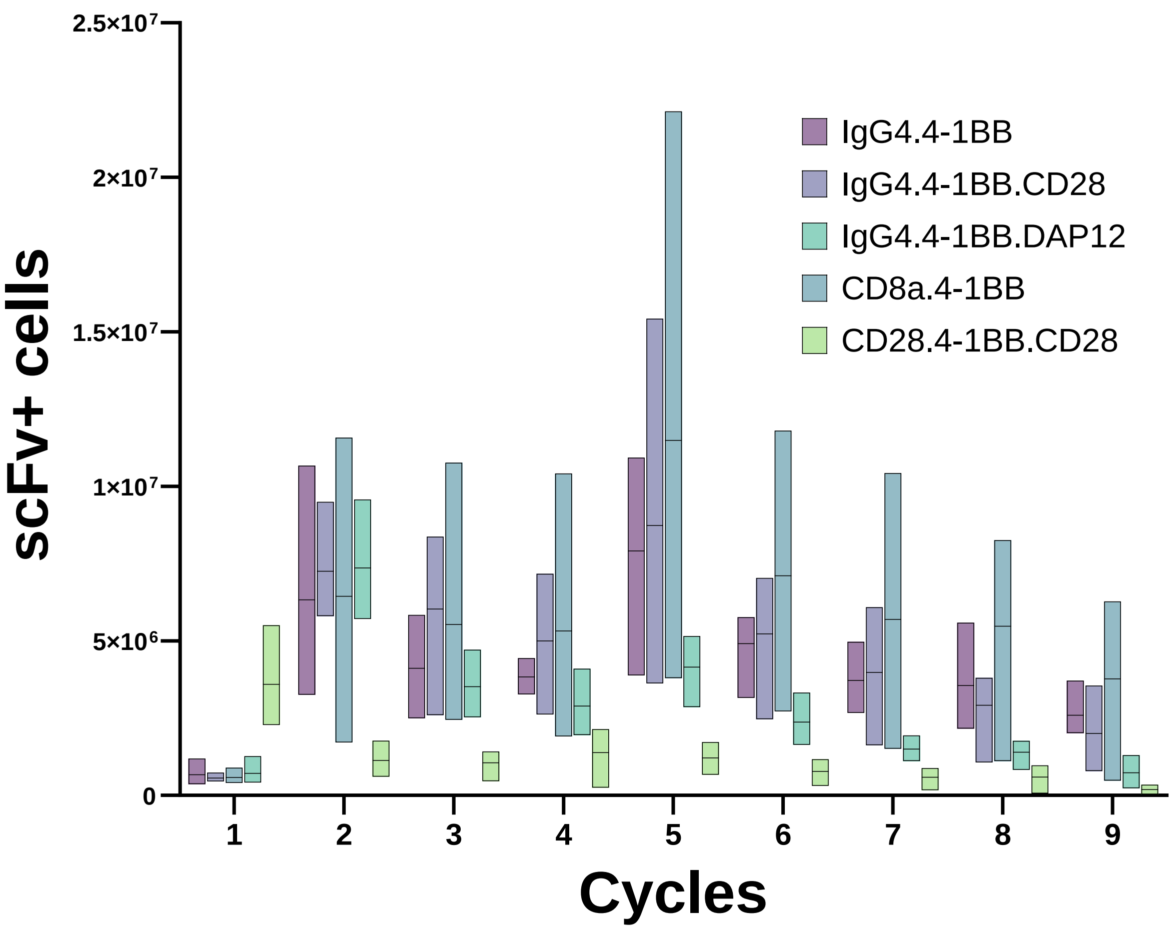


Figure S9. CD19-CARs expansion after repeated antigen stimulation.
T cells were transduced with the different CD19-CAR constructs and surviving CAR+ cells were measured via flow cytometry after repeat antigen stimulation, adding Raji cells at a 1:1 initial ratio every 2-3 days.

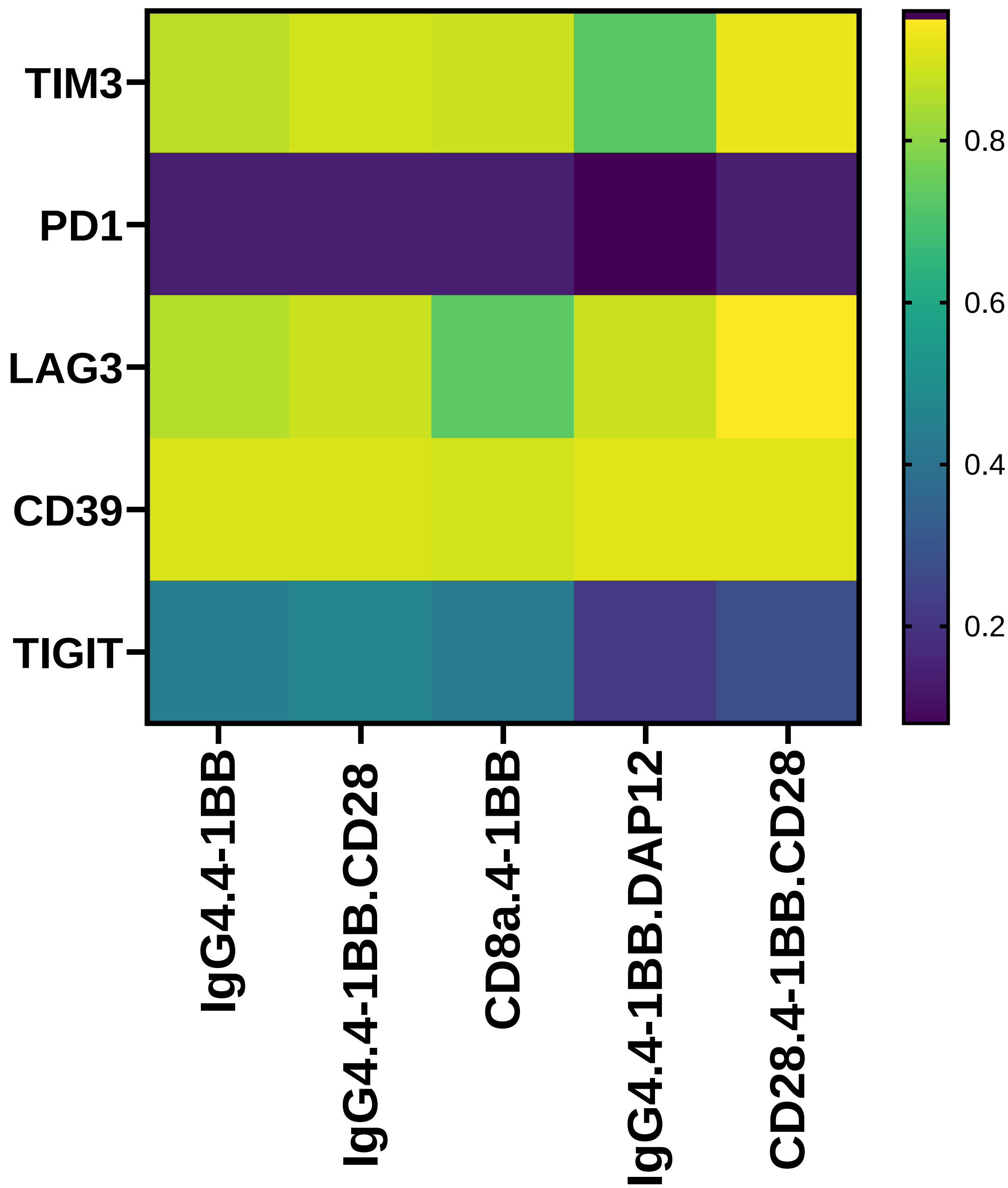


Figure S10. *In vitro* characterization of CD19-CAR constructs exhaustion markers
T cells were transduced with the different CD19-CAR constructs and exposed to 11 cycles of repeat tumor antigen stimulation, then exhaustion markers were measured via flow cytometry.

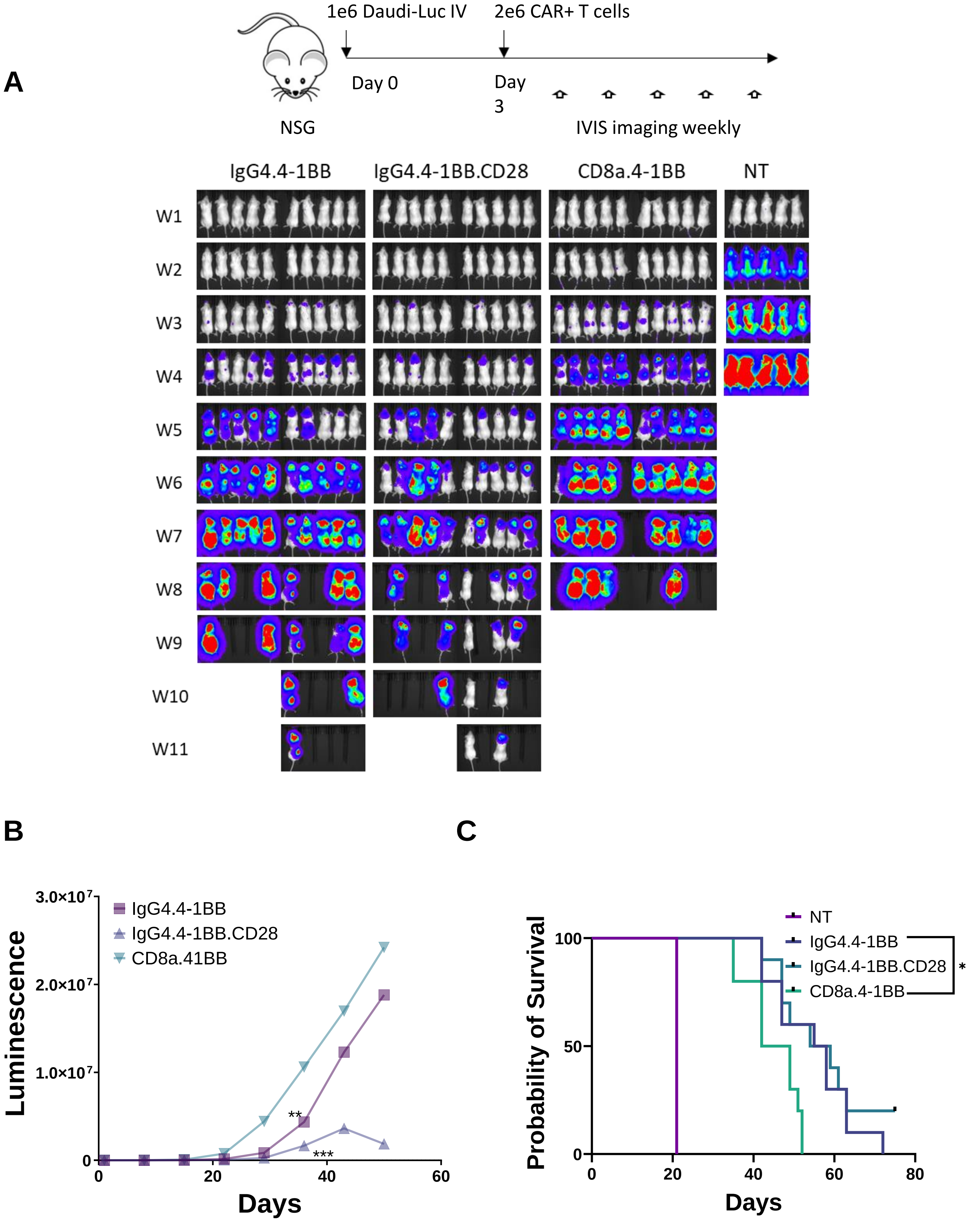

A
1e6 Daudi-Luc IV
2e6 CAR+ T cells
Day 0
Day 3
NSG
IVIS imaging weekly
IgG4.4-1BB
IgG4.4-1BB.CD28
CD8a.4-1BB
NT
W1
W2
W3
W4
W5
W6
W7
W8
W9
W10
W11
B
Luminescence
3.0×10^7
2.0×10^7
1.0×10^7
0
IgG4.4-1BB
IgG4.4-1BB.CD28
CD8a.41BB
**
***
0
20
40
60
Days
C
Probability of Survival
100
50
0
NT
IgG4.4-1BB
IgG4.4-1BB.CD28
CD8a.4-1BB
*
0
20
40
60
80
Days

Figure S11. *In vivo* characterization of CD19-CAR constructs with second donor. NSG mice were injected intravenously with $1x10^6$ luciferase-labeled CD19+ Daudi cells, then three days later received $2x10^6$ CAR+ cells. Tumor growth was monitored by weekly bioluminescence imaging. (A-B) Monitoring tumor bioluminescence demonstrates both IgG4.4-1BB and IgG4.4-1BB.CD28 significantly reduce tumor progression, *** $p \leq 0.001$; ** $p \leq 0.01$ two-way ANOVA with Turkey's multiple comparison. (C) Kaplan-Meier curve generated from survival of animals in (A); ** $p \leq 0.01$, *$p \leq 0.05$ (log-rank Mantel-Cox test).

Table S1. Sequences of Domains used in library.

GD2 scFv	MEFGLSWLFLVAILKGVQCSRDILLTQTPLSLPVSLGDQASISCRSSQSLVHRNGNTYLHWYLQKPGQSPKLLIHKVSNRFSGVPDRFSGSGSGTDFTLKISRVEAEDLGVYFCSQSTHVPPLTFGAGTKLELKRADAAPTVSIFPGSGGGGSGGEVKLQQSGPSLVEPGASVMISCKASGSSFTGYNMNWVRQNIGKSLEWIGAIDPYYGGTSYNQKFKGRATLTVDKSSSTAYMHLKSLTSEDSAVYYCVSGMEYWGQGTSVTVSSTR
CD19 scFv	MEFGLSWLFLVAILKGVQCSRDIQMTQTTSSLSASLGDRVTISCRASQDISKYLNWYQQKPDGTVKLLIYHTSRLHSGVPSRFSGSGSGTDYSLTISNLEQEDIATYFCQQGNTLPYTFGGGTKLELKRGGGGSGGGGSGGGGSGGGGSEVQLQQSGPGLVAPSQSLSVTCTVSGVSLPDYGVSWIRQPPRKGLEWLGVIWGSETTYYNSALKSRLTIIKDNSKSQVFLKMNSLQTDDTAIYYCAKHYYYGGSYAMDYWGQGTTVTVSSYVTVSSQDPA
CD8a Hinge	TTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACD
IgG4 Hinge	ESKYGPPCPSCP
CD28 Hinge	IEVMYPPPYLDNEKSNGTIIHVKGKHLCPSPLFPGPSKP
CD8a TM	IYIWAPLAGTCGVLLLSLVIT*LYC*
$CD28_{wt}$ Costim	RSKRSRLLHSDYMNMTPRRPGPTRKHYQPYAPPRDFAAYRS
$CD28_{mt}$ Costim	RSKRSRLLHSDFMNMTPRRPGPTRKHYQAYAAPRDFAAYRS
TNFRSF9(4-1BB) Costim	KRGRKKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCEL
TNFRSF4 Costim	ALYLLRRDQRLPPDAHKPPGGGSFRTPIQEEQADAHSTLAKI
DAP12 Costim	YFLGRLVPRGRGAAEAATRKQRITETESPYQELQGQRSDVYSDLNTQRPYYK
$CD3\zeta_{wt}$	RVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPQRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
$CD3\zeta_{mt}$	RVKFSRSADAPAYQQGQNQLFNELNLGRREEFDVLDKRRGRDPEMGGKPQRRKNPQEGLFNELQKDKMAEAFSEIGMKGERRRGKGHDGLFQGLSTATKDTFDALHMQALPPR

Table S2. Sequences of Domains used in library.

RPL32 Reverse Primer	GGGCAGTTGCATCTTCATATT
RPL32 Forward Primer	CAAGGAAAGACGAGCTGTAGG
GD2 Reverse Primer	TAGACAAATCGTCCAGCACAG
GD2 Forward Primer	CCAGTACTCCATTCCGCTTAC